SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING COURSE
CODE: CSA4305

COURSE OUTCOME COVERED

CO-2 : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool : Short Answer Test

Weightage: 10%

QUESTIONS: **Total Marks: 20**

Question 1 (5 Marks)

Suppose that a company wants its website to work seamlessly on mobile, tablet, and desktop devices. Explain the concept of responsive web design and its importance. Describe the key CSS techniques used to achieve responsiveness, and explain how CSS Flexbox or Grid can be used to design adaptive layouts. Illustrate your answer with a simple example layout and suggest improvements to enhance user experience across different devices.

Question 2 (5 Marks)

Suppose that a webpage contains a form where user input must be validated before submission. Explain the concept of client-side validation and its purpose. Describe how JavaScript event handling is used in validation and explain the role of regular expressions in validating inputs such as email or phone numbers. Provide a suitable example and suggest improvements for better usability and security.

Question 3 (5 Marks)

Suppose that a webpage layout breaks when the browser window is resized. Explain the CSS box model and its components, and describe how CSS positioning techniques affect layout structure. Identify possible causes of layout breaking and explain how media queries can be used to resolve these issues. Suggest best practices to ensure a stable and responsive layout.

Question 4 (5 Marks)

Suppose that a website requires dynamic content updates without reloading the entire page. Explain the concept of the Document Object Model (DOM) and its role in web development. Describe how JavaScript can be used to manipulate the DOM for updating webpage content dynamically. Provide an example and suggest improvements to enhance performance and user experience.

Code of Conduct:

I N.Loka Bandhava/192324167 ___ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response

CO2 - AT 3.

N. Loka Bandhava
192324167

①

Responsive web Design (RWD)

Responsive web Design is a web development approach that allows a website to automatically adjust its layout and content according to different screen size such as mobiles phones, tables and desktops. It ensures that users have a consistent user-friendly experience on any device.

Importance :-

-> provides a better user experience on all devices

-> Improves websites accessibility.

-> Reduce the need to create seperate websites for mobile and desktop.

-> Makes website maintaince easier.

css Technique for Responsiveness

1. Flexible layouts

```
.container {
    width : 80%;
}
```

2. Flexible Images

```
img {
    max-width : 100%;
    height : auto;
}
```

3. Media Queries :-

```
@media (max-width : 768px) {
    .container {
        width: 100%; }}
```

using CSS Flexbox

```
.container {
display : flex;
flex-wrap : wrap;
}
.box {
flex : 1;
Padding: 20px;
}
```

Example layout :-

Desktop

Header

Menu | content | sidebar

Footer

Mobile

Header

Menu

Content

Sidebar

footer.

Improvements :

- use readable font sizes
- Make buttons large enough for touch screens
- optimize image for faster loading.

Conclusion :-

Responsive web Design ensures that websites are attractive, fuctionaly, and easy to use on every device.

② Client - side validation

Client - side validation checks the user's input in the browser before sending the data to the server.

Purpose :-

- → prevents invalid data entry
- → gives immediate feedback to users.
- → Improves user experience
- → Reduces unnecessary server requests.

Java script event Handling:

Java script validates form data using events such as.

- onsubmit
- on change
- onkeyup
- onblur.

Example :-

```
<form onsubmit = "return validate Form ()">
<input type =" email" id= " email">
<input type = " submit">
</form>
<script>
function validate form () {
let email = document. get Element ById ("email").value;
if (email == "") {
alert ("Email is required");
return false;
}
return true;
}
</script>
```

Regular expression :-

Regular expression (Regex) are patterns used to verify weather user input matches a required format.

Email validation ex:-

let pattern = /^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]

Phone number validation ex:-

let phone = /^[0-9]{10}$/;

Improvements:-

-> Display clear error messages
-> Highly incorrect fields
-> validate while typing.
-> use HTML5 validation attributes like required and Pattern

Conclusion :-

Client-side validation improves usability and reduces errors. java script events and regular expression help validate user input efficiently.

③ CSS Box Model:

Every HTML elemented is treated as a rectangular box.

The Box models has four parts.

1. Content - Actual text or image
2. padding - Space around the content
3. Border - surrounds the padding
4. Margin - space outside the border.

CSS positioning Techniques

- Static – Default position
- Relative – Moves Relative to its normal position.
- Absolute – positioned relative to its nearest positioned parent.
- Fixed – Remains fixed on the screen.
- Sticky – changes b/w relative and fixed based on scrolling.

Causes of Layout Breaking:

- fixed width element
- Large images without responsive sizing.
- Improper use of positioning.
- missing media queries.
- overflow caused by long content.

Media Queries:

```
@media (max-width : 768 px) {
    .Container {
    display : flex;
    flex- direction : column;
    }
}
```

Best practices

→ use percentage widths instead of fixed pixels.
→ use flexbox or CSS grid.
→ Apply media queries for different screen sizes.
→ use box-sizing: border-box
→ Make images responsive.
→ Test layouts on multiple devices and browsers.

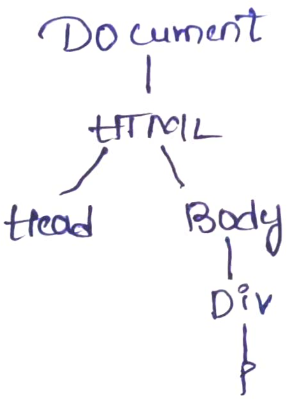
Conclusion:-

The CSS box model, positioning techniques, and media queries help create responsive layouts that remain stable even when browser size changes.

(4) **Document Object Model (DOM) :-**

The Document object Model (DOM) is a programming interface that represents an HTML Document as a tree of objects. Each HTML element becomes an object that Java Script can access and modify.

Example of DOM Tree :



Role of DOM

- Access HTML elements
- Modify webpage content
- changes style dynamically
- Add or remove elements
- Respond to user actions.

Java Script DOM Manipulation

```
<h2 id = "tittle"> welcome </h2>
<button onclick ="changes Text ()">
click Here
</button>
<script>
function change Text (){
document. get Element By Id ("tittle"). inner HTML=
"welcome to Internet programming";
}
</script>
```

Improvements

- use efficient DOM methods like query selector ()
- Minimize unnecessary DOM updates
- Use event delegation for better performance
- Load Javascript after the HTML content.

Conclusion:-

The DOM enables Java script to dynamically update webpage content, creating interactive, fast, and user-friendly websites without requiring a full page reload.